

UGINO, Russian Federation

WMO: 310160

Lat: 58.633N

Lon: 128.500E

Elev: 231

StdP: 98.59

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-48.4	-46.1	-49.9	0.0	-48.2	-47.7	0.0	-45.9	3.4	-21.7	3.0	-23.3	0.4	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	29.8	19.4	27.6	18.2	25.6	17.2	20.7	26.7	19.6	25.3	18.4	24.0	1.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.8	23.4	17.5	12.9	22.2	16.4	12.0	21.1	61.1	26.6	57.2	25.2	53.4	24.0	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.3	3.4	2.8	DB	-50.3	33.6	3.6	2.3	-52.9	35.2	-55.1	36.6	-57.1	37.9	-59.8	39.5	(4)
(5)			WB	-50.1	23.3	3.8	1.9	-52.8	24.7	-55.0	25.8	-57.2	26.9	-59.9	28.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-6.4	-34.0	-28.0	-15.1	-2.8	7.1	15.1	17.9	15.4	6.5	-5.3	-23.0	-31.5	(6)
(7)		DBStd	19.41	7.72	7.08	7.31	5.52	4.21	3.65	3.44	3.61	4.06	5.89	7.65	8.20	(7)
(8)		HDD10.0	6574	1363	1063	777	384	109	5	0	4	118	473	990	1287	(8)
(9)		HDD18.3	9072	1621	1296	1036	634	349	107	51	104	356	732	1240	1546	(9)
(10)		CDD10.0	603	0	0	0	0	18	159	244	171	12	0	0	0	(10)
(11)		CDD18.3	60	0	0	0	0	0	11	37	13	0	0	0	0	(11)
(12)		CDH23.3	959	0	0	0	0	14	252	505	186	2	0	0	0	(12)
(13)		CDH26.7	276	0	0	0	0	2	53	179	42	1	0	0	0	(13)
(14)	Wind	WSAvg	1.0	0.6	0.8	1.0	1.3	1.3	1.1	1.0	0.9	0.9	1.1	0.8	0.7	(14)
(15)	Precipitation	PrecAvg	504	18	12	15	25	51	61	85	77	67	45	28	21	(15)
(16)		PrecMax	642	33	21	31	61	80	113	152	150	126	102	64	37	(16)
(17)		PrecMin	397	7	5	7	10	23	26	30	17	23	22	11	12	(17)
(18)		PrecStd	68	6	4	6	12	15	21	29	31	22	17	12	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-12.3	-7.3	6.7	15.3	25.0	31.0	33.8	30.8	21.9	10.6	3.0	-5.6	(19)
(20)		MCWB	-13.2	-8.2	1.6	7.6	13.4	19.0	21.0	19.8	13.8	5.5	0.8	-6.8	(20)	
(21)		2%	DB	-16.7	-10.8	2.2	11.3	21.6	28.1	31.1	27.8	18.7	6.9	-3.9	-10.8	(21)
(22)		MCWB	-17.3	-11.5	-0.9	4.7	11.6	17.5	20.1	18.7	12.0	3.7	-5.6	-11.3	(22)	
(23)		5%	DB	-19.6	-13.9	-0.2	9.1	18.8	26.3	28.7	25.5	16.0	4.2	-8.9	-15.7	(23)
(24)		MCWB	-19.9	-14.6	-2.6	3.3	10.2	16.4	19.2	17.8	10.8	1.9	-9.6	-16.1	(24)	
(25)		10%	DB	-22.6	-17.0	-2.7	7.0	16.4	24.6	26.2	23.3	13.6	2.3	-12.2	-20.0	(25)
(26)		MCWB	-22.9	-17.6	-4.6	2.3	8.8	15.7	18.2	17.0	9.5	0.7	-12.7	-20.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-13.0	-8.0	1.9	7.8	14.2	20.7	22.6	21.2	15.0	6.1	0.4	-6.6	(27)
(28)		MCDWB	-12.2	-7.0	5.5	15.2	23.3	26.2	30.7	27.5	19.1	9.7	2.7	-5.6	(28)	
(29)		2%	WB	-17.2	-11.4	-0.7	5.1	12.4	18.7	21.2	20.0	13.0	4.4	-5.2	-11.3	(29)
(30)		MCDWB	-16.8	-10.8	1.6	10.5	20.1	25.3	27.9	25.1	17.0	6.7	-4.0	-10.8	(30)	
(31)		5%	WB	-19.9	-14.4	-2.6	3.5	10.9	17.6	20.2	18.7	11.5	2.2	-9.6	-16.0	(31)
(32)		MCDWB	-19.5	-13.8	-0.3	8.5	16.9	24.6	26.3	23.9	15.2	3.8	-8.9	-15.7	(32)	
(33)		10%	WB	-22.8	-17.5	-4.7	2.3	9.5	16.5	19.2	17.7	10.0	0.7	-12.7	-20.2	(33)
(34)		MCDWB	-22.6	-17.0	-2.7	6.6	15.2	22.9	25.0	22.7	13.0	2.2	-12.2	-20.0	(34)	
(35)	Mean Daily Temperature Range	MDBR	9.0	13.3	17.4	15.0	13.5	14.4	13.3	13.0	11.2	8.5	9.7	8.3	(35)	
(36)		5% DB	MCDBR	9.9	14.5	16.7	16.6	19.4	18.7	18.5	17.2	16.8	9.5	10.9	10.4	(36)
(37)		MCWBR	9.8	13.8	13.8	10.6	10.8	9.6	9.2	9.0	10.4	6.3	9.9	10.1	(37)	
(38)		5% WB	MCDBR	10.0	14.4	16.1	16.0	17.1	16.0	14.8	14.6	14.6	8.7	10.7	10.4	(38)
(39)		MCWBR	9.8	13.7	13.3	10.4	10.1	9.1	8.3	8.5	9.5	6.1	9.7	10.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.246	0.282	0.318	0.393	0.466	0.438	0.479	0.430	0.349	0.308	0.256	0.229	(40)	
(41)		taud	2.006	2.027	2.013	1.925	1.981	2.213	2.113	2.239	2.416	2.240	2.027	1.948	(41)	
(42)		Ebn at Noon	554	719	803	795	763	799	750	763	786	693	527	433	(42)	
(43)		Edh at Noon	54	88	120	158	161	130	141	116	83	71	51	40	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.64	1.67	3.40	5.09	4.81	5.49	5.06	3.99	2.51	1.30	0.80	0.39	(44)	
(45)		RadStd	0.04	0.10	0.13	0.31	0.41	0.39	0.38	0.33	0.20	0.14	0.05	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.84	N/A	N/A	N/A	N/A

Nomenclature: See separate page